Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI automation builder with 8+ years shipping production systems at enterprise scale -- track record of embedding with operational teams to map workflow bottlenecks, prototype generative AI solutions, and transform proofs-of-concept into 0-to-1 production-grade AI automations with security controls, observability, and HITL checkpoints. At Entrata, eliminated 80% of manual operational workload via automated pipelines for 100K+ monthly active housing units. I build and evaluate agentic systems using LangChain, LangGraph, and MCP: AgentProbe ships a 5-metric LLM-as-judge evaluation framework to ensure AI safety and quantify business impact; Leasel ships an 8-agent LangGraph pipeline with HITL escalation and audit logging on regulated compliance decisions. M.S. Data Analytics Engineering, GMU (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Workflow Automation & Agentic Pipelines | LangChain & LangGraph | Model Context Protocol (MCP) | AI Evaluation Frameworks | Human-in-the-Loop (HITL) | Prompt Engineering | SQL Analytics | Stakeholder Adoption & Enablement

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building ML models and AI pipelines integrating data systems with back-end APIs and reporting layers; debugging data-driven features across the application stack.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern

- **Turned a fully manual survey-request workflow into an end-to-end agentic automation** using a LangGraph stateful graph (parse -> plan -> retrieve -> validate -> conditional routing) -- delivering grounded, risk-aware compliance decisions automatically.
- Implemented a RAG layer (retrieval over FAA Part 107 + operational corpus) with validation guardrails enforcing altitude limits, battery budgets, and no-fly-zone checks -- blocking non-compliant missions before export.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- **Demystified AI tooling and drove adoption across 600+ students** -- authored internal guides and best practices to translate complex generative AI and database concepts for non-technical learners; managed stakeholder ecosystem across multiple sections and faculty leads.
- Built multi-source prediction pipelines (Python, R) with R2 = 0.977 regression accuracy; applied Isolation Forest for automated data integrity checks; designed Tableau/Quarto dashboards for faculty leadership (+40% research comprehension).
- Fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; presented model design and evaluation results to faculty.

Entrata (PropTech / Real Estate SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Embedded with operational teams to identify workflow bottlenecks and eliminate 80% of manual workload** -- built Python/Bash automation pipelines for income-limit imports, gross-rent updates, and document workflows across 100K+ monthly active housing units.
- **Built and owned the RabbitMQ-based async gross-rent processing pipeline** -- designed retry logic and idempotency guarantees eliminating double-processing failures; zero compliance violations across 7 years of production release cycles on regulated HUD/LIHTC data.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, and stored procedure refactoring -- used SQL to surface data correctness issues in eligibility and financial workflows before production.
- Led AWS S3 document migration with security controls -- 100% PHI compliance via Libsodium-encrypted pipelines, zero security incidents; shipped 20+ production modules across 120+ Agile sprints; quantified business impact through SQL-driven adoption metrics across eligibility and financial workflows.
- Mentored 5+ engineers; authored architectural recipes and best practices via 300+ code reviews; built internal debugging tools reducing team debugging time by 40%.

PROJECTS

Leasel: Agentic AI Compliance Investigation Platform

LangGraph 8-agent -- HITL -- FastAPI -- TF-IDF RAG

Three-surface enterprise AI knowledge platform: **Document Intelligence** (OCR-fallback ingestion, hybrid TF-IDF retrieval, citation grounding, human-review gate on regulated decisions); **Support Investigation** (8-agent LangGraph StateGraph with HITL escalation when confidence too low, conditional routing across Triage/Retrieval/Config/History/Reasoning/Validation agents); **Tenant Self-Service** (eligibility questions always route to staff, never resolved by AI). Core rule: LLM reasons; deterministic logic or human makes every final compliance call. Results: 93.6% precision / 87.9% F1 on 1,240 lease PDFs; 100% citation correctness on Phase 1 eval set.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: Open-Source AI Evaluation Toolkit LangGraph -- LangChain -- FastAPI -- ChromaDB -- MCP

Established an end-to-end evaluation framework for AI safety and performance: 15-endpoint FastAPI backend, 5-metric LLM-as-judge engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI/CD regression harness that automatically blocks failing agent runs, multi-provider LLM Router (OpenAI/Anthropic/Groq/Ollama), offline + online evaluation loops. Used to evaluate LeaseI across 1,240+ compliance test cases. 100% local, zero vendor cost.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: NLP-Driven Clinical Records Query System Flask -- LLMs -- BERT -- React -- Docker

Reduced manual clinical retrieval time ~60% by building an NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; HIPAA-compliant RBAC across structured medical records; drove staff adoption of AI tooling across clinical workflows.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur 2017

SKILLS

AI & Agents: LangChain, LangGraph, MCP (Model Context Protocol), Generative AI, Agentic AI, Multi-Agent Orchestration, HITL, LLM Evaluation, LLM-as-Judge, Evaluation Frameworks, Prompt Engineering, RAG, ChromaDB, Embeddings, LLM Routing, Gemini Models, OpenAI API, Anthropic API, FastAPI, Observability, Guardrails, Hallucination Detection

Languages & Frameworks: Python, JavaScript, TypeScript, Node.js, Java, SQL, PL/SQL, PHP, R, React.js, Django, Spring Boot

Cloud & Infra: AWS (S3, EC2, Lambda, Glue, Bedrock), Azure AI Foundry, Azure OpenAI, Vertex AI, GCP, Docker, Kubernetes, GitHub Actions CI/CD, PostgreSQL, RabbitMQ, Redis

ML & Data: PyTorch, scikit-learn, XGBoost, Random Forest, Isolation Forest, Pandas, NumPy, Tableau, Power BI